Predicting Football Player Market Value Using LASSO and Regression Trees

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Introduction

Football player market values vary substantially across players and may depend on a combination of age, technical ability, physical attributes, and other performance characteristics. The goal of this analysis is to investigate which player attributes are most useful for predicting market value and whether allowing for nonlinear relationships improves predictive performance.

The analysis uses male player data from EA Sports FC 24. The natural logarithm of market value, in euros, is used as the response variable, while 32 characteristics describing age, body size, attacking ability, technical skill, movement, power, mentality, and defending are considered as candidate predictors. Two statistical learning methods are compared: LASSO regression and regression trees. LASSO provides a relatively interpretable linear model with automatic variable selection, while regression trees can capture nonlinear relationships and interactions among player characteristics.

The primary objectives are to identify attributes associated with player market value and to compare the out-of-sample predictive performance of the two methods.

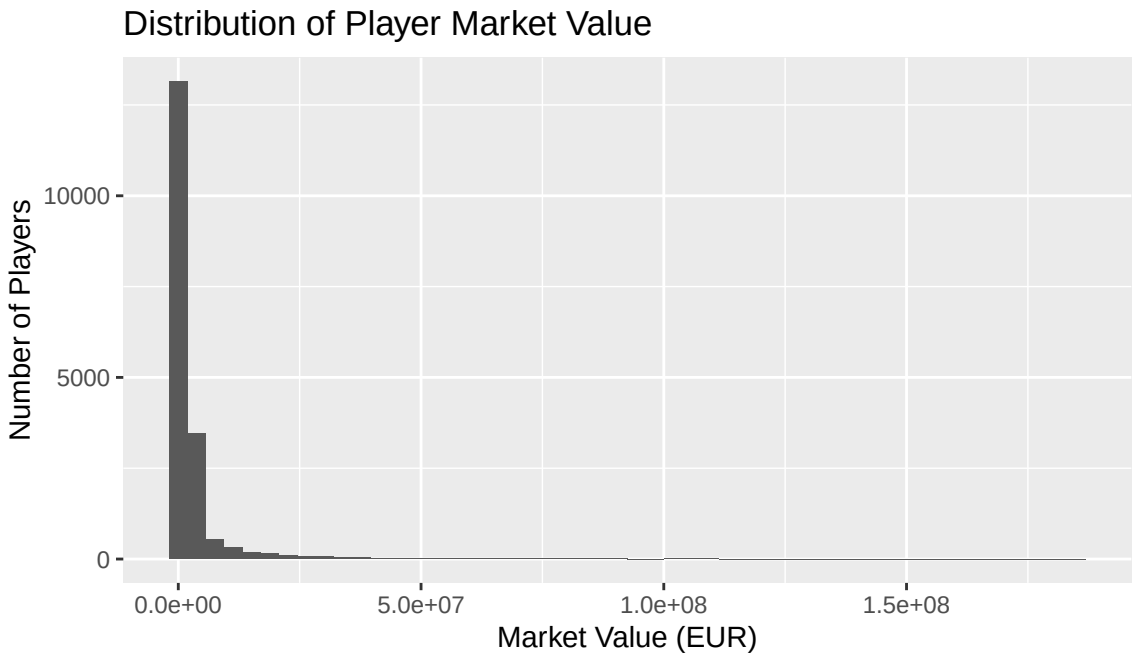
Data Preparation

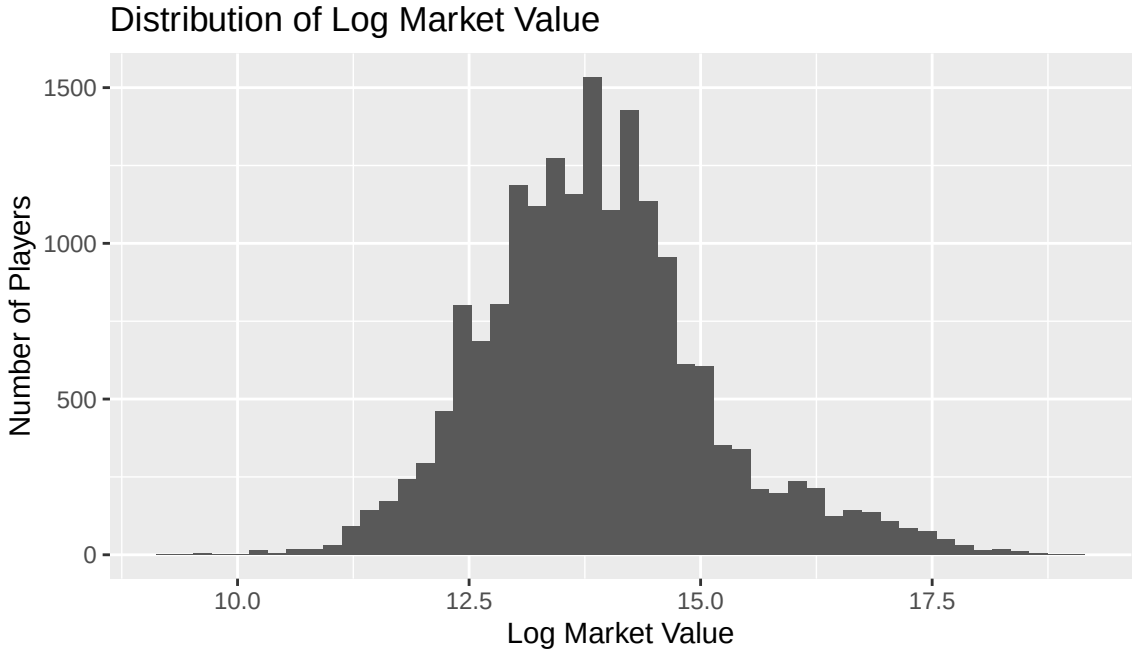
The data were obtained from the *EA Sports FC 24 Complete Player Dataset* published on Kaggle by Stefano Leone. The dataset contains player information from multiple FIFA/EA Sports FC releases. The analysis was restricted to male players from FC 24, resulting in 18,350 observations. Market value was recorded in euros and contained 100 missing observations.

Because market value was extremely right-skewed, with a relatively small number of highly valuable players, the natural logarithm of market value was used as the response variable. The log transformation produced a substantially more symmetric distribution and reduced the influence of extreme market values.

For modelling, goalkeepers (GK) were excluded because their relevant skills differ substantially from those of players in other positions. After removing the 100 observations with missing market values and excluding goalkeepers, the resulting modelling dataset contained 16,217 players and 32 candidate predictors.

Variables closely related to EA's overall valuation system, including overall rating, potential, wage, and release clause, were excluded to avoid making the prediction problem too artificially easy. Aggregate ratings such as pace, shooting, passing, dribbling, defending, and physical were also omitted in favor of their more detailed underlying attributes. This allowed the models to identify which individual player characteristics contained useful predictive information.





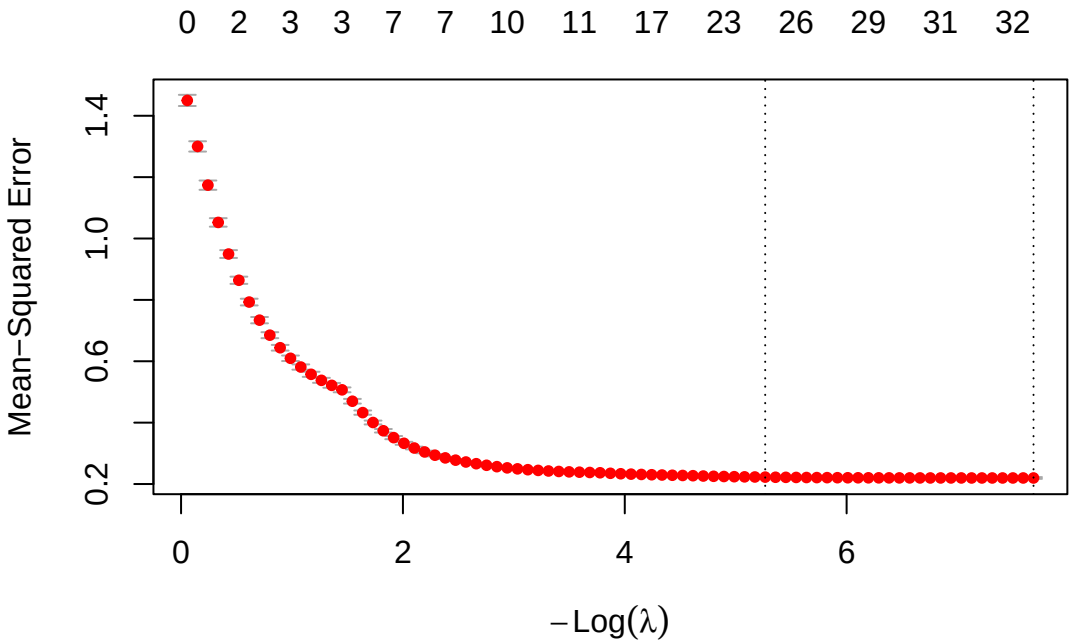
The data were randomly divided into an 80% training set and a 20% test set using a fixed random seed for reproducibility. This resulted in 12,973 training observations and 3,244 test observations. Model selection and tuning were performed using the training data, while the test data were reserved for the final comparison of predictive performance.

LASSO Regression

LASSO regression was first used to predict log market value. The method adds an L_1 penalty to the regression objective, which shrinks coefficient estimates and can set some coefficients exactly equal to zero.

This provides a form of variable selection and is useful here because many of the player attributes may contain overlapping information.

The tuning parameter λ was selected using cross-validation on the training set. Although $\lambda_{\min} = 0.000460$ produced the minimum cross-validation error, it retained all 32 predictors. The one-standard-error choice, $\lambda_{1SE} = 0.00516$, was therefore used because it gives a simpler model whose cross-validation error remains within one standard error of the minimum.



At the one-standard-error value of λ, LASSO retained 25 of the 32 candidate predictors. Among the retained variables were age, movement reactions, ball control, short passing, and composure. Age had a negative coefficient, indicating that, conditional on the other predictors in the model, older players tended to have lower predicted log market values. Movement reactions, ball control, short passing, and composure had positive coefficients.

Seven variables were shrunk to zero: height, volleys, curve, free-kick accuracy, aggression, interceptions, and sliding tackle. Their exclusion does not necessarily imply that these attributes have no relationship with market value; rather, they did not provide sufficient additional predictive information given the other variables included in the model.

On the held-out test set, the LASSO model achieved an RMSE of 0.4741 on the log market value scale. This value will be used as the benchmark for comparison with the regression tree.

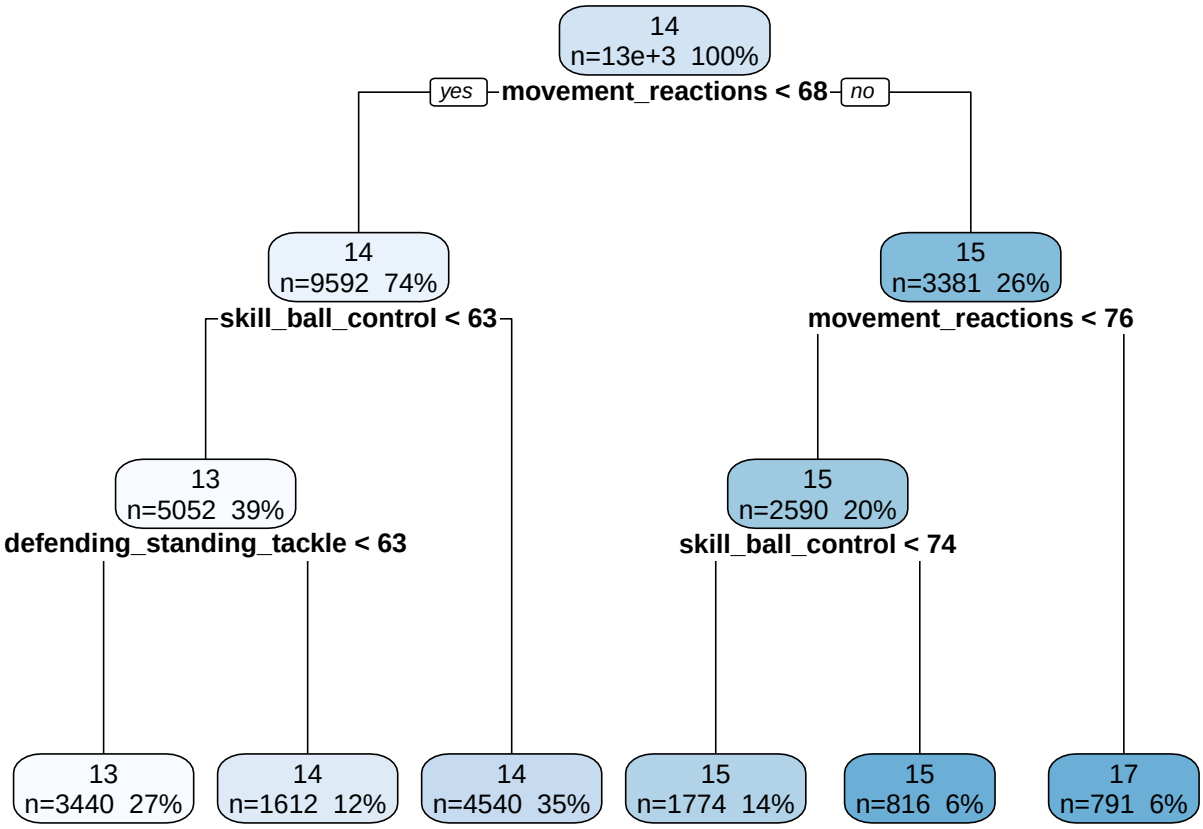
Regression Tree

A regression tree was also fitted to allow for nonlinear relationships and interactions among player attributes. Unlike LASSO, which assumes an additive linear relationship between the predictors and log market value, a regression tree recursively divides players into groups based on predictor thresholds.

A large initial tree was grown using the training data and then cost-complexity pruning was considered using cross-validation. The one-standard-error criterion selected a complexity parameter of approximately

0.000134, corresponding to a tree with 274 splits. Although this tree was used for prediction, its size makes it impractical to visualize and interpret directly.

To illustrate the main structure of the fitted relationship, a smaller tree was constructed for visualization using a larger complexity parameter. This simplified tree is presented only for interpretation; predictive performance is evaluated using the cross-validation-selected tree described above.



The first split in the simplified tree occurs at a movement reactions rating of 68, indicating that reactions provide a particularly strong separation in predicted market value. Reactions appear again at a threshold of 76 among players with higher reaction ratings. Ball control is also used prominently, while standing tackle provides an additional split among players with lower reactions and ball control.

These results are broadly consistent with the LASSO model, which also retained movement reactions and ball control with positive coefficients. The agreement between two substantially different modelling approaches suggests that these attributes contain useful predictive information for player market value.

The cross-validation-selected regression tree achieved a test RMSE of 0.4188, compared with 0.4741 for LASSO. Thus, the regression tree reduced test RMSE by approximately 11.7%. This suggests that nonlinear relationships and interactions among player characteristics provide additional predictive information beyond the additive linear structure used by LASSO.

Model	Test RMSE
LASSO	0.4741
Regression Tree	0.4188

Discussion and Conclusion

This analysis compared LASSO regression and regression trees for predicting the market value of male outfield players in EA Sports FC 24. Both methods identified meaningful structure in the data, but they emphasized different aspects of the prediction problem.

LASSO produced a relatively interpretable model and reduced the predictor set from 32 variables to 25 using the one-standard-error rule. Age was negatively associated with predicted log market value, while attributes such as movement reactions, ball control, short passing, and composure were positively associated with value. The fact that several variables were shrunk to zero suggests that some player attributes contribute little additional predictive information once correlated characteristics are already included.

The regression tree achieved better out-of-sample predictive performance, with a test RMSE of 0.4188 compared with 0.4741 for LASSO, representing an improvement of approximately 11.7%. The first split of the simplified tree was based on movement reactions, with ball control and standing tackle also appearing in later splits. The repeated importance of movement reactions and ball control across both methods suggests that these variables are particularly useful for predicting player market value.

The stronger performance of the regression tree suggests that market value may depend on nonlinear relationships and interactions that are not fully captured by an additive linear model. However, the tree selected by cross-validation was very large, with 274 splits, which reduces its interpretability. LASSO therefore offers a simpler global summary, while the regression tree provides stronger predictive accuracy.

There are several limitations to this analysis. The market values are generated within the EA Sports FC database and should not be treated as exact real-world transfer valuations. In addition, variables such as league, club, nationality, contract status, and player position were not included, even though they may contain additional information about market value. Future work could incorporate these variables or compare the current models with alternative methods such as ridge regression or more flexible tree-based approaches.

References

Leone, S. *EA Sports FC 24 Complete Player Dataset*. Kaggle. Accessed August 2026.

Click here for the link: EA Sports FC 24 Complete Player Dataset